Quiz 3 Solution

Q1. Which one of these is a VALID application of Union-Find?

a) Predicting the temperature distribution in a heat exchanger.
b) Storing hierarchical data like a tree structure.
c) Determining if fluid flow in a network of pipes creates a loop.
d) Measuring the voltage drop across a series of resistors.

Correct: c)

Q2. Which Data Structure is MOST efficient if M union operations are to be performed on N elements?

a) Quick-Find.
b) Quick-Union.
c) Weighted Quick-Find
d) None.

Correct: d)

Q3. M union and find operations, on a set of N elements using Weighted Quick-Union will have ____ time complexity?

a) O(M*N)
b) O(N)
c) O(log(N))
d) O(Mlog(N))

Correct: d)

Q4. What is the key difference between Quick-Union and Weighted Quick-Union?

a) Quick-Union uses an array to store parent pointers, while Weighted Quick-Union uses a tree structure
b) Quick-Union does union operation in O(N), while Weighted Quick-Union does union operation in O(NlogN)
c) Quick-Union uses extra space, while Weighted Quick-Union does not
d) Weighted Quick-Union ensures balanced trees, while Quick-Union does not

Correct Answer: d)

Q5. In the Quick-Union implementation of Union-Find, what is the time complexity of the find(p)?

a) O(Log(N))
b) O(1)
c) O(N)
d) O(NlogN)

Correct: c)

Q6. Given the following parent array representation of a disjoint set and a union operation, what will be the final array if Quick-Union is used?

Notation: union(p, q) --> p's Leader becomes q's Leader.

	0	1	2	3	4	5	6	7	8	9
parent[]	0	1	1	2	3	5	6	7	8	9

- *union(8, 4)*

a)

0	1	2	3	4	5	6	7	8	9
0	8	1	2	3	5	6	7	8	9

b)

0	1	2	3	4	5	6	7	8	9
0	1	1	2	3	5	6	7	1	9

c)

0	1	2	3	4	5	6	7	8	9
0	1	1	2	3	5	6	7	3	9

d)

0	1	2	3	4	5	6	7	8	9
0	8	8	2	3	5	6	7	8	9

Correct: a)

Q7. Which of the following is TRUE about the depth of trees in Weighted Quick-Union?

a) The depth of any tree is at most $\log_2 N$.
b) The depth of any tree is at most N.
c) The depth of any tree is at most 1.
d) The depth of any tree is at most $\log_{10} N$
e) Correct: a)

Q8. Which code snippet correctly implements union function of Weighted Quick-Union data structure from Union-Find.

```
void union(int p,int q){
    int root1 = find(p);
    int root2 = find(q);

    if(root1 == root2) return;

    else if(size[root1] > size[root2]){
        parent[root2] = root1;
        size[root1] += size[root2];
    }
    else{
        parent[root1] = root2;
        size[root2] += size[root1];
    }
}
```

a)

```
void union(int p,int q){
    int root1 = find(p);
    int root2 = find(q);

    if(root1 == root2) return;

    if(size[root1] >= size[root2]){
        int temp = root1; root1 = root2; root2 = temp;
    }
    parent[root1] = root2;
    size[root2] += size[root1];
}
```

b)

a) only a
b) only b
c) both
d) none

Correct: c)

Q9. In Weighted Quick-Union, what is the purpose of keeping track of the size of each tree?

a) To balance the trees and keep them shallow
b) To ensure the same size across all trees
c) To reduce the space complexity
d) To sort the elements in each tree

Correct: a)

Q10. Given the following parent array representation of a disjoint set and a union operation, what will be the final array if Weighted Quick-Union is used?

	0	1	2	3	4	5	6	7	8	9
parent[]	0	9	9	3	4	4	6	7	8	9

- *union(5,2)*

a)

0	1	2	3	4	5	6	7	8	9
0	9	9	3	9	4	6	7	8	9

b)

0	1	2	3	4	5	6	7	8	9
0	9	9	3	4	4	6	7	8	4

c)

0	1	2	3	4	5	6	7	8	9
0	9	4	3	4	4	6	7	8	9

d)

0	1	2	3	4	5	6	7	8	9
0	9	9	3	4	9	6	7	8	9

Correct: a)

11. What will be the output of the following code?

```
class A {
public:
    A() { cout << "A "; }
    A(int x) { cout << "A(int) "; }
};
class B : public A {
public:
    B() { cout << "B "; }
};
int main() {
    B obj;
    return 0;
}
```

a) A(int) B

b) A B

c) Compilation error

d) B A

Correct Answer: b)

12. In the Quick-Find implementation of Union-Find, what is the time complexity of the union(p,q)?

a) O(1)

b) O(log N)

c) O(N)

d) O(N log N)

Correct Answer: c)

13. Which of the following statements about multiple inheritance in C++ is true?

a) A class can inherit from multiple classes using a comma-separated list.

b) C++ does not support multiple inheritance.

c) A derived class can only inherit from one base class.

d) Multiple inheritance requires virtual functions to work.

Correct Answer: a)

14. Consider the following parent array representation of a disjoint set:

Index: 0 1 2 3 4 5 6 7

Parent: [0 1 2 3 4 5 6 7]

Using quick-find algorithm, If we perform the operations union(3, 4) the parent array is

Parent: [0 1 2 3 3 5 6 7]

followed by union(4, 6), what will be the updated parent array?

a) [0, 1, 2, 6, 4, 5, 6, 7]

b) [0, 1, 2, 3, 3, 5, 4, 7]

c) [0, 1, 2, 3, 3, 5, 3, 7]

d) [0, 1, 2, 4, 4, 5, 4, 7]

Correct Answer: c)

15. In the basic Union-Find data structure, what happens during the 'union' operation?

a) It connects two disjoint sets into one by modifying parent pointers.

b) It merges two sets by sorting their elements.

c) It reduces the number of elements in each set by half.

d) It assigns the same rank to all elements in a set.

Correct Answer: a)

16. What is the default access specifier for members of a class in C++?

a) Public

b) Private

c) Protected

d) Default access specifiers do not exist in C++

Correct Answer: b)

17. What happens if a class contains a private constructor but no friend functions?

a) Objects of the class cannot be created directly outside the class.

b) Objects of the class can still be created using the `new` keyword.

c) The program will throw a compilation error.

d) The class cannot have member functions.

Correct Answer: a)

18. Which of the following correctly describes the role of a base class constructor in inheritance?

a) The base class constructor is never called when a derived class object is created.

b) The default constructor of the base class is automatically called when a derived class object is created.

c) A base class constructor must always be explicitly called inside the derived class constructor.

d) The base class constructor is always called after the derived class constructor.

Correct Answer: b)

19. Which of the following can we use the find operation of union-find to do?

a) It determines whether two elements belong to the same set.

b) It retrieves the smallest element in a set.

c) It sorts all elements within a set.

d) It deletes an element from a set.

Correct Answer: a)

20. Which of the following correctly describes the 'diamond problem' in C++?

a) A compilation error that occurs when a class has two destructors.

b) A scenario where multiple inheritance causes ambiguity due to repeated inheritance of the same base class.

c) A memory leak caused by improper use of dynamic memory allocation.

d) A runtime error caused by an infinite recursion in function calls.

Correct Answer: b)